Texus Hackathon

NILAS-Nilgiris Intelligent Liquid waste Automated System

PROBLEM STATEMENT:

The Nilgiris the In rugged, steep terrain makes centralized sewerage systems impossible. Current open drains often block or leak, causing human waste to overflow into pristine mountain springs and the Bhavani River. Manual monitoring is dangerous and slow, leading to environmental contamination and health risks for indigenous communities.

SOLUTION:

An automated, real-time "Digital Shield" using IoT and Long Range Wide AreaNetwork technology. Smart sensor nodes installed in manholes detect blockages and rising levels instantly, sending alerts to a centralized dashboard before contamination occurs.

TECHNICAL FRAMEWORK:

Controller: ESP32 (chosen for its deep-sleep/low-power capabilities).

Connectivity: LoRaWAN (SX1278). Unlike Wi-Fi or GSM, it penetrates thick forest cover and reaches 5-15 km across hilly valleys without needing a SIM card.

Sensing: JSN-SR04T Waterproof Ultrasonic Sensor to measure depth and Flow Sensors to pinpoint leakage locations.

Power: Solar-powered with Li-ion battery backup for 24/7 autonomous operation.

DATA VISUALIZATION:

Workflow: Sensor → LoRa Gateway → The Things Network (Cloud) → Live Heatmap Dashboard.

The Heatmap: A color-coded map (Green/Yellow/Red) that allows municipal officers to visualize the entire drainage network's health at a glance.

Alerts: Automatic WhatsApp/SMS notifications with GPS coordinates when a "Critical" level is reached.

HARDWARE RUGGEDIZATION:

To survive "Sewer Gas" (H_{2}S) and the Nilgiris humidity:

PCB: Protected with Conformal Coating (Silicone) to prevent corrosion.

Enclosure: IP67/NEMA 4X Polycarbonate boxes.

Sealing: Hermetic gaskets and IP68 cable glands to keep toxic gases out.

SDG DOMAINS:

1.CLEAN WATER AND SANITATION(6)

PROTECTING HEADWATER

2.SUSTAINABLE CITIES(11)

HILL-STATON INFRASTRUCTURE

3.LIFE ON LAND(15)

KEEPING FOREST AREA CLEAN AND CHEMICAL FREE

SIMULATION OUTPUT:

Instead of just numbers, present your simulation as a Digital Twin—a virtual model of the actual drainage network in the Nilgiris.

1. The Scenario: "The Monsoon Surge"

Show a 24-hour time-lapse simulation.

Variable: 150mm of rainfall over 4 hours.

The Data: Your simulation should show water levels rising across 15 different "virtual nodes" placed at strategic points near the Bhavani River headwaters.

2. Visual Output: The Heatmap Dashboard

Present a screen-recorded or live-click mock-up of the dashboard.

Normal State: All icons are Green. Flow rate is 0.5 m/s.

Anomaly Detection: Node #04 (near a steep ravine) turns Yellow. The system identifies a "Slow Drain" pattern—this suggests a physical obstruction (like plastic or silt) rather than just rain.

Critical State: Node #04 turns Red. The simulation triggers an automated logic gate.

📲The "Actionable Intelligence" Output

The most important part of your simulation is the end result. What does the human on the other side see?

The Automated SMS/WhatsApp Payload

Show a mock-up of the alert sent to the Municipal Engineer's phone:

⚠ NILAS CRITICAL ALERT

Location: Kadavoor Sector 4 (Spring-Side)

GPS: 11.3214° N, 76.9322° E

Status: 90% Capacity - Overflow Imminent

Impact Risk: High (Bhavani River Headwaters)

Action: Dispatch Team Beta for Manual Desilting.

📊 The "Idea-Centric" Simulation Metrics

To impress the judges, include a "Simulation Summary" table that proves the efficiency of your specific technical choices (LoRaWAN + ESP32):

GRAFANA AND PROMETHEUS

To implement Prometheus and Grafana for your NILAS project, you aren't just making a graph; you are building an industrial-grade observability stack.

In a hackathon, using these tools proves you can handle Time-Series Data—which is critical for predicting when a manhole in Kadavoor will overflow before it actually happens.

1. The Architecture: "The Data Bridge"

Since LoRaWAN nodes send data to The Things Network (TTN), and Prometheus likes to "pull" data from a web endpoint, you need a bridge.

The Setup: Use Node-RED as your bridge. It has a "TTN Node" to receive your sensor data and a "Prometheus Exporter" node to host that data for Prometheus.

2. Designing the "Digital Twin" Dashboard in Grafana

Since your path is idea-centric, your dashboard should be the "Star" of the show. Use these specific panels:The Geomap Panel: Use the Geomap plugin to place your nodes on a satellite map of the Nilgiris. Set the marker color to change based on the water level (Green < 50\%, Yellow 70\%, Red > 90\%).The Gauge Panel: For each critical manhole, show a "Water Level Gauge."The "Rate of Change" Stat:

PromQL query :

Rate(manhole-leve-[5m])

This shows how fast the water is rising. If the rate is high, even if the level is low, your "Digital Twin" can trigger an early warning.